

Coupling is a through-bond effect

Do neighbouring nuclei interact through space or through the electrons in the bonds? We know that coupling is in fact a ‘through-bond effect’ because of the way coupling constants vary with the shape of the molecule. The most important case occurs when the protons are at either end of a double bond. If the two hydrogens are *cis*, the coupling constant J is typically about 10 Hz, but if they are *trans*, J is much larger, usually 15–18 Hz. These two chloro acids are good examples.

If coupling were through space, the nearer *cis* hydrogens would have the larger J . In fact, coupling occurs *through the bonds* and the more perfect parallel alignment of the bonds in the *trans* compound provides better communication and a larger J .

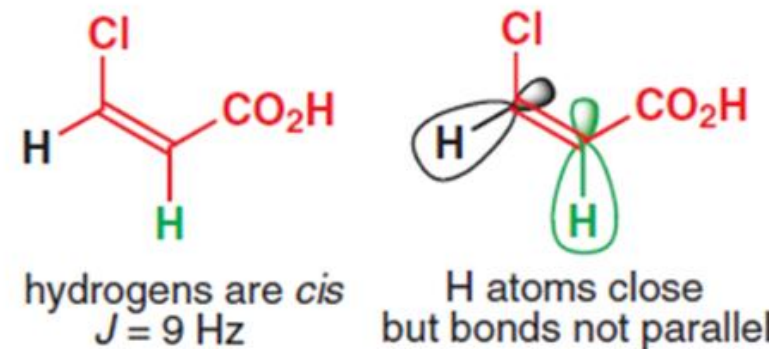
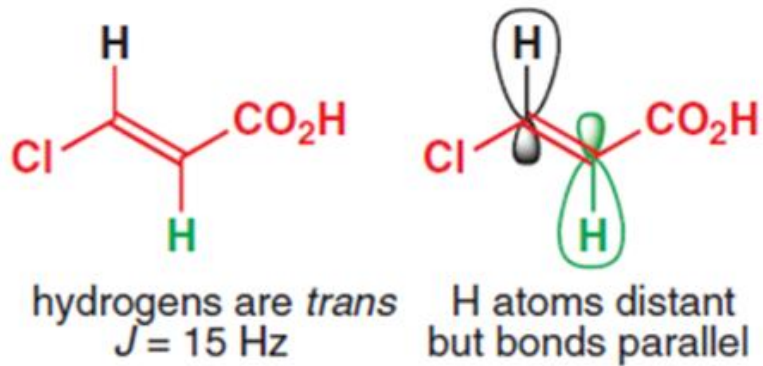


Table of Characteristic Proton NMR Shifts

<i>type of proton</i>	<i>type of compound</i>	<i>chemical shift range, ppm</i>
RCH_3	1° aliphatic	0.9
R_2CH_2	2° aliphatic	1.3
R_3CH	3° aliphatic	1.5
$\text{C}=\text{C}-\text{H}$	vinyllic	4.6-5.9
$\text{C}=\text{C}-\text{H}$	vinyllic, conjugated	5.5-7.5
$\text{C}\equiv\text{C}-\text{H}$	acetylenic	2-3
$\text{Ar}-\text{H}$	aromatic	6-8.5
$\text{Ar}-\text{C}-\text{H}$	benzylic	2.2-3
$\text{C}=\text{C}-\text{CH}_3$	allylic	1.7
$\text{HC}-\text{F}$	fluorides	4-4.5
$\text{HC}-\text{Cl}$	chlorides	3-4
$\text{HC}-\text{Br}$	bromides	2.5-4

HC-Br	bromides	2.5-4
HC-I	iodides	2-4
HC-OH	alcohols	3.4-4
HC-OR	ethers	3.3-4
RCOO-CH	esters	3.7-4.1
HC-COOR	esters	2-2.2
HC-COOH	acids	2-2.6
HC-C=O	carbonyl compounds	2-2.7
RCHO	aldehydic	9-10
ROH	hydroxylic	2-4
ArOH	phenolic	4-12
C=C-OH	enolic	15-17
RCOOH	carboxylic	10.0-13.2
RNH ₂	amino	1-5

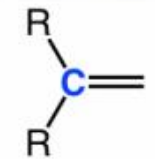
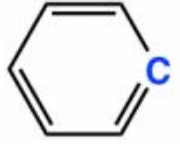
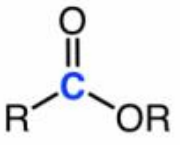
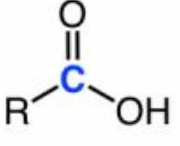
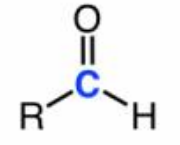
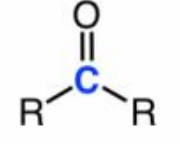
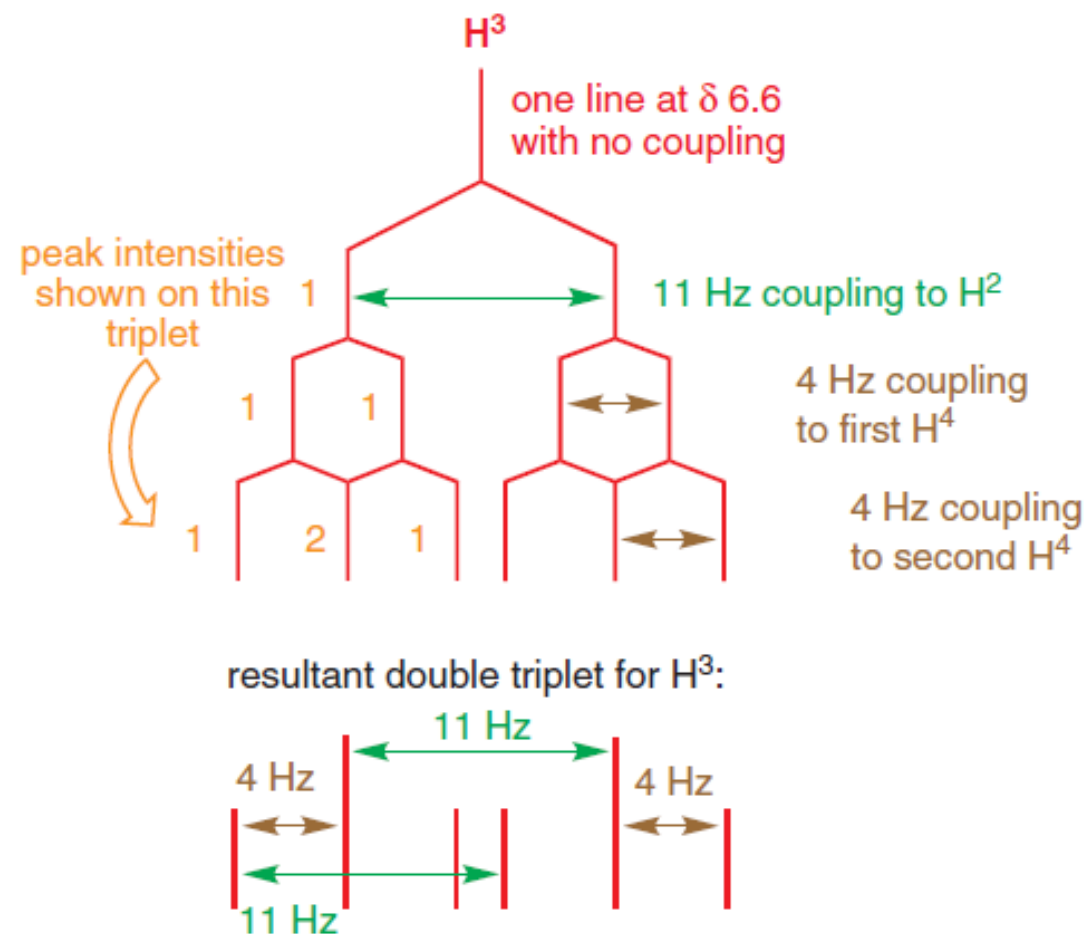
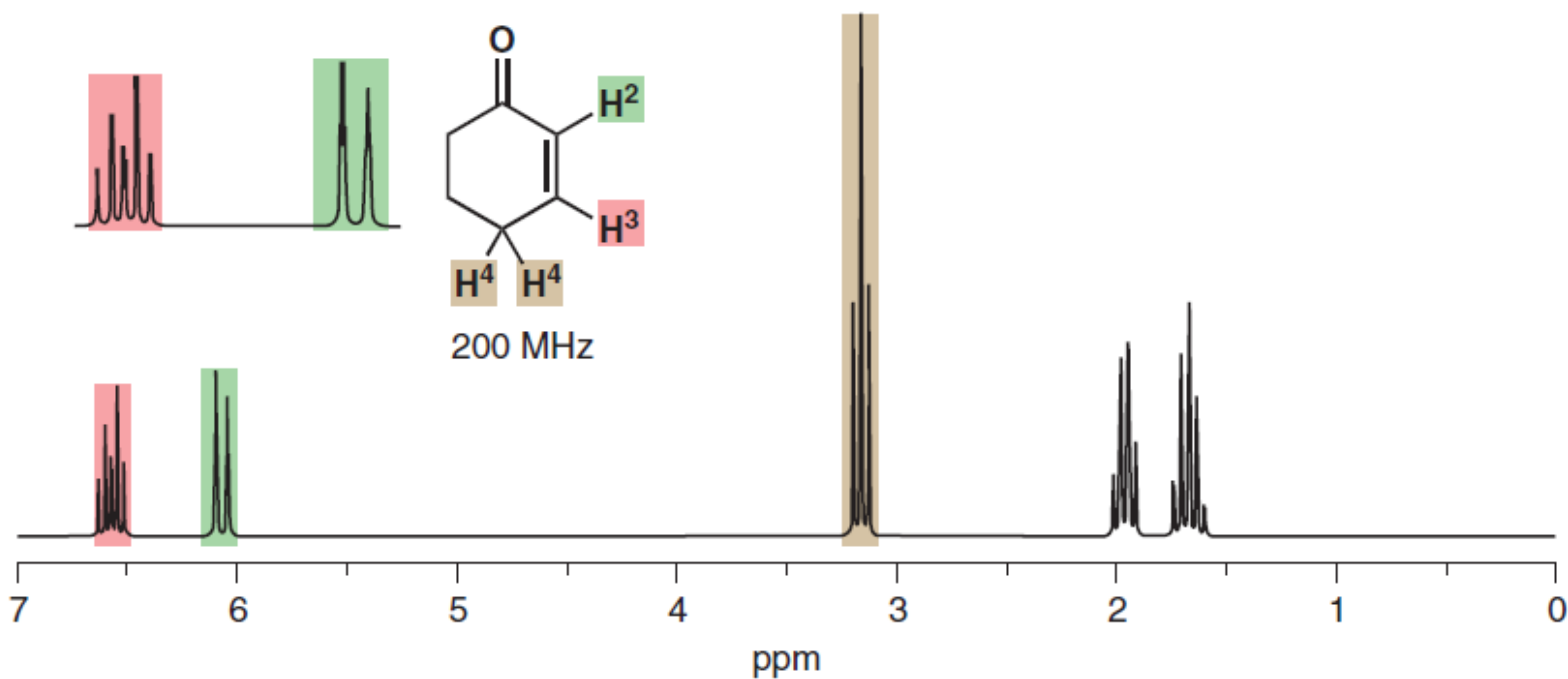
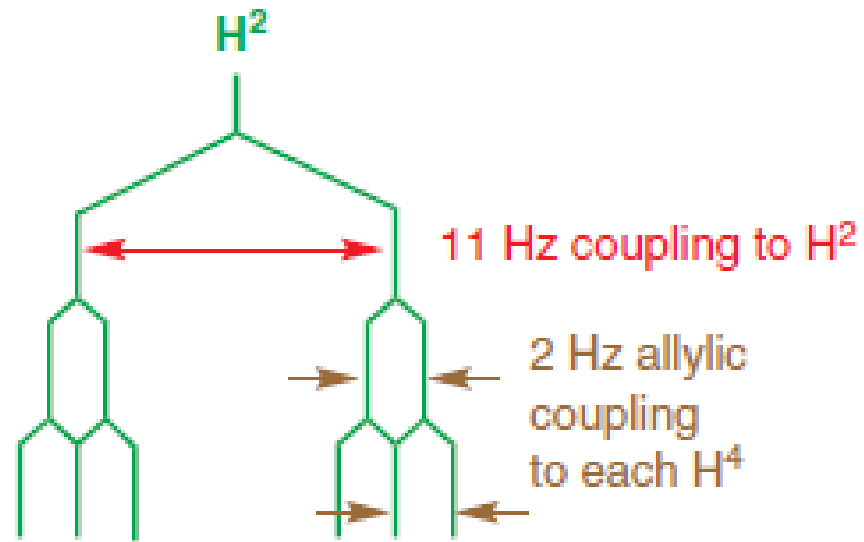
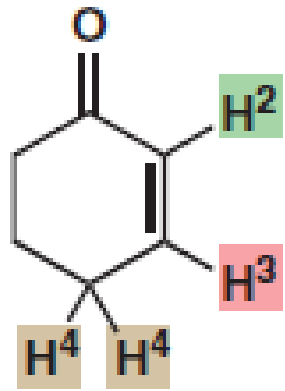
Type of Carbon	Chemical Shift (ppm)	Type of Carbon	Chemical Shift (ppm)
$\text{R}-\text{CH}_3$	0 – 35		80 – 150
$\text{R}-\text{CH}_2\text{R}$	15 – 55		110 – 170
$\text{R}-\text{CH}(\text{R})\text{R}$	25 – 55		165 – 175
$\text{R}-\text{C}(\text{R})_3$	30 – 40		175 – 185
—C—X (X: Cl, Br or N)	10 – 65		190 – 200
—C—O—	50 – 90		200 – 220
$\text{R}-\text{C}\equiv$	70 – 90		

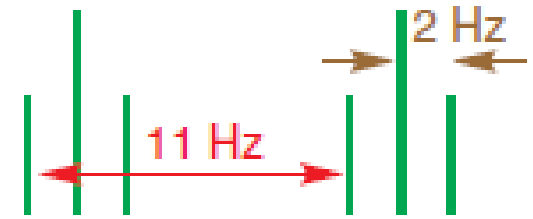
Table 6.3 Approximate ^{13}C NMR chemical shifts of some common groups

Coupling is at least as helpful as chemical shift in assigning spectra. The proton next to the carbonyl group (H² in the diagram) has one neighbour (H³) and appears as a doublet with $J = 11$ Hz, just right for a proton on a double bond with a *cis* neighbour. The proton H³ itself appears as a double triplet. Inside each triplet the separation of the lines is 4 Hz and the two triplets are 11 Hz apart.





resultant double triplet for H^2 :



Homotopic, Enantiotopic and Diastereotopic Nuclei

Homotopic nuclei

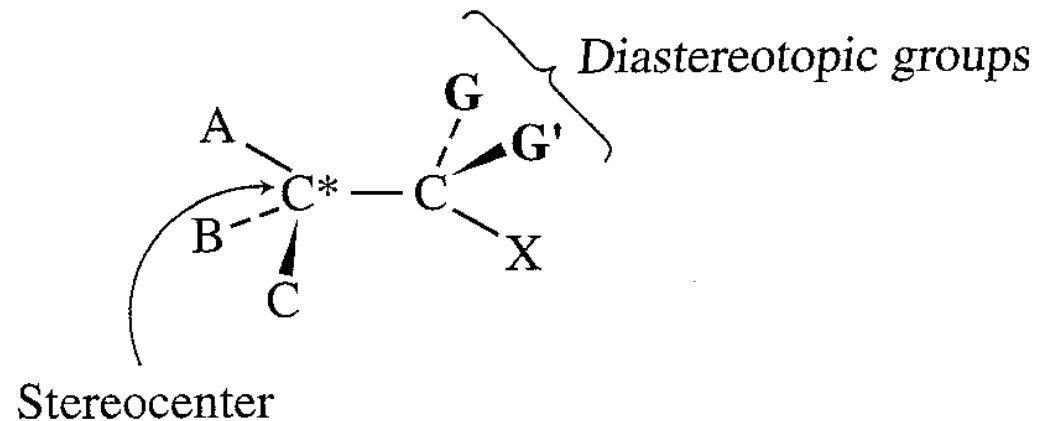
- Homotopic groups are always equivalent. They are related to each other either by *a bond rotation or an axis of rotation in the molecule*
- In the absence of coupling from other groups, they are isochronous and give a single NMR absorption

Enantiotopic nuclei

- Enantiotopic groups are equivalent, are typically isochronous and give a single NMR absorption unless they are placed in a chiral environment
- Related to each other by a reflective symmetry element (the most common being a mirror plane within the molecule)

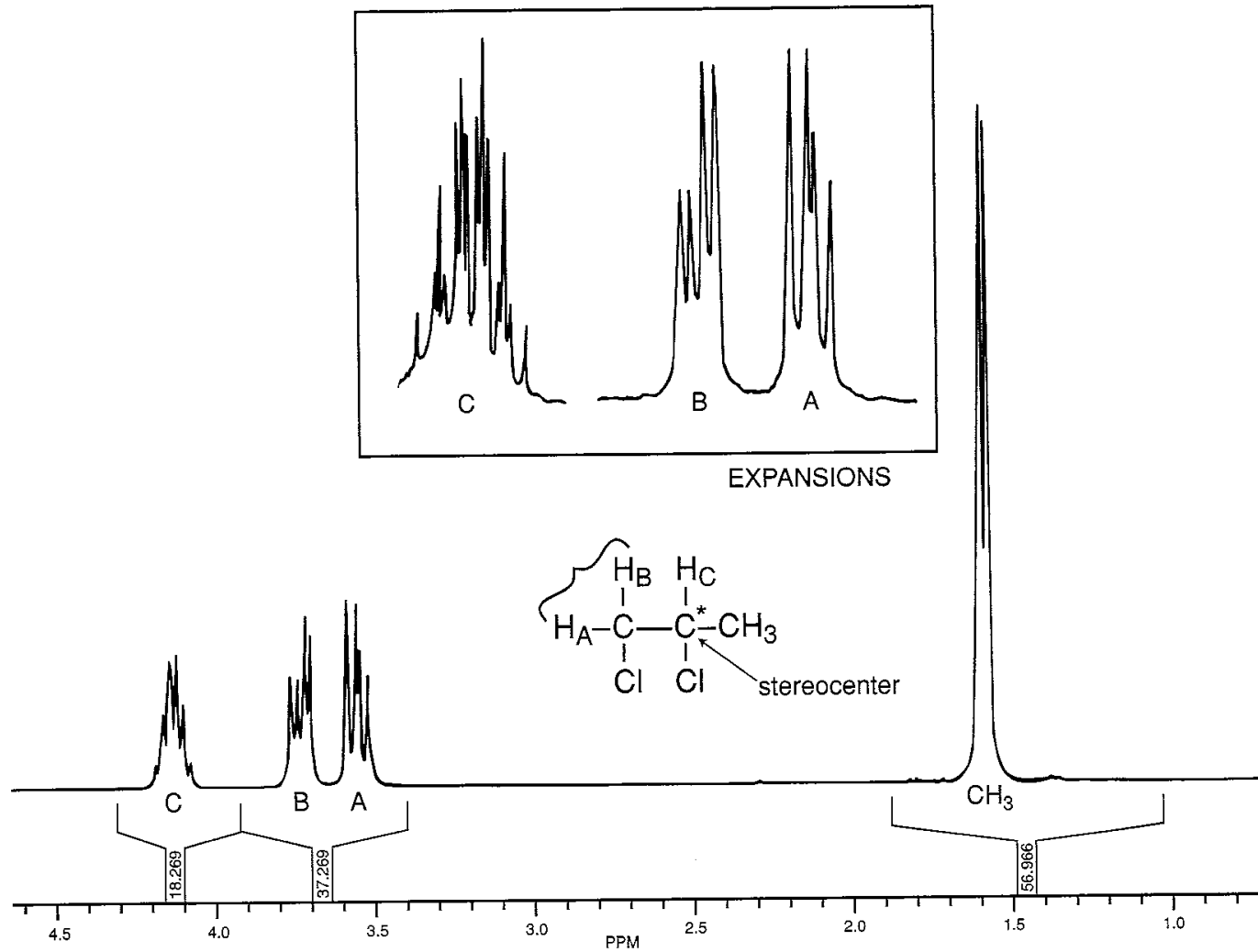
Diastereotopic groups

- They are not equivalent, not isochronous
- Different chemical shifts in NMR
- When the diastereotopic groups are hydrogens, they often split with a coupling constant 2J



In many cases, the difference in chemical shifts of diastereotopic hydrogens is so small that neither this difference nor the coupling is easily detectable

Diastereotopic Protons

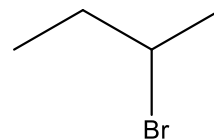
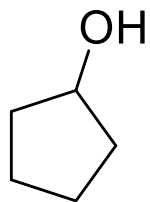
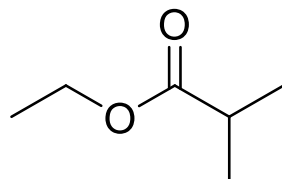


Problem Set 1

6. A compound with molecular formula $C_3H_6O_2$ has two peaks (both singlet) in the proton NMR spectra. Predict the structure.

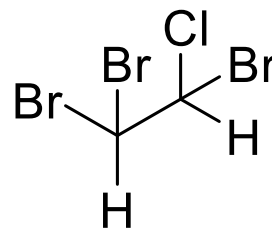
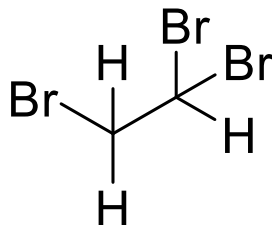
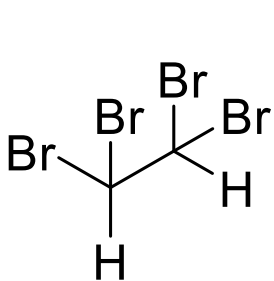
1H : 2.19 (3H), 3.66 (3H)

5. Predict the no. of proton NMR peak will be observed in each of the molecule:



1. In the 400 MHz ^1H NMR spectrum of an organic compound, a doublet is observed. The two lines of the doublet appear at δ 2.35 and 2.38 ppm. Calculate the coupling constant (J) value.

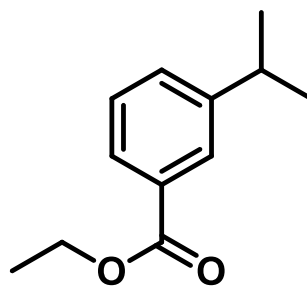
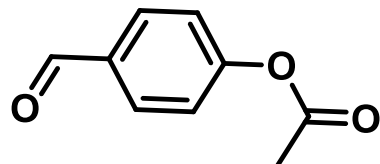
2. Write down the no. of different types of proton present in each molecule and then show their splitting.



3. ^1H NMR spectrum of an organic compound recorded on a 500 MHz spectrometer showed a quartet with line positions at 1759, 1753, 1747, 1741 Hz. Find the chemical shift (δ) and coupling constant (Hz) of the quartet.

(CSIR NET)

4. Draw the NMR of the compounds with approximate δ value in correct order with labelling and proper splitting.



5. An organic compound ($C_{10}H_{12}O_2$) exhibits the following 1H NMR spectra data. Predict the structure

δ 7.80 (2H, d, $J = 8$ Hz), 6.80 (2H, d, $J = 8$ Hz), 4.10 (2H, q, $J = 7.2$ Hz), 2.4 (3H, s), 1.25 (3H, t, $J = 7.2$ Hz)

(GATE)

C is the number of Carbon atoms

H is the number of Hydrogen

X is the number of Halogen (F, Cl, Br, I)

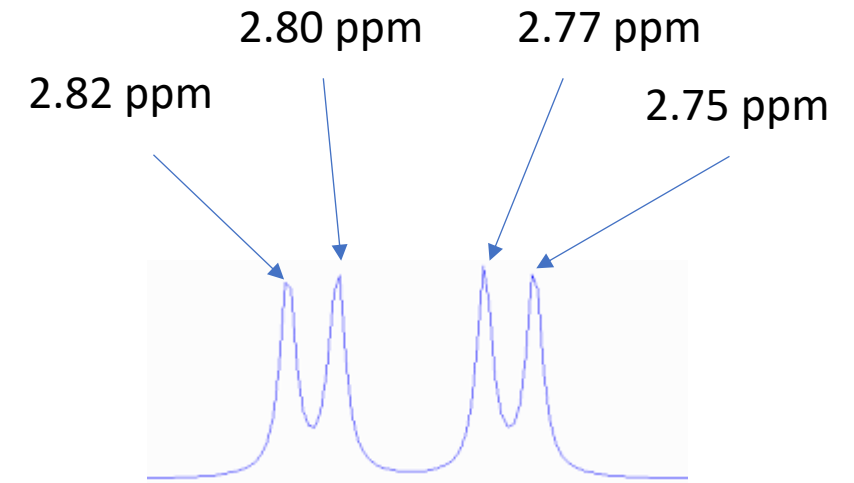
N is the number of Nitrogen.

Oxygen and sulfur are not included here

$$DBE = C + 1 - \frac{H}{2} - \frac{X}{2} + \frac{N}{2}$$

•Alkene $\rightarrow$ DBE = 1 Alkyne $\rightarrow$ DBE = 2 Benzene $\rightarrow$ DBE = 4 (3 double bonds + 1 ring)

6. One proton peak split in doublet of doublet (as shown) due to presence of two types of neighbouring proton. Calculate the coupling constants.
Given that the instrument frequency is 400 MHz. The delta value of each peak
Shown by arrow.

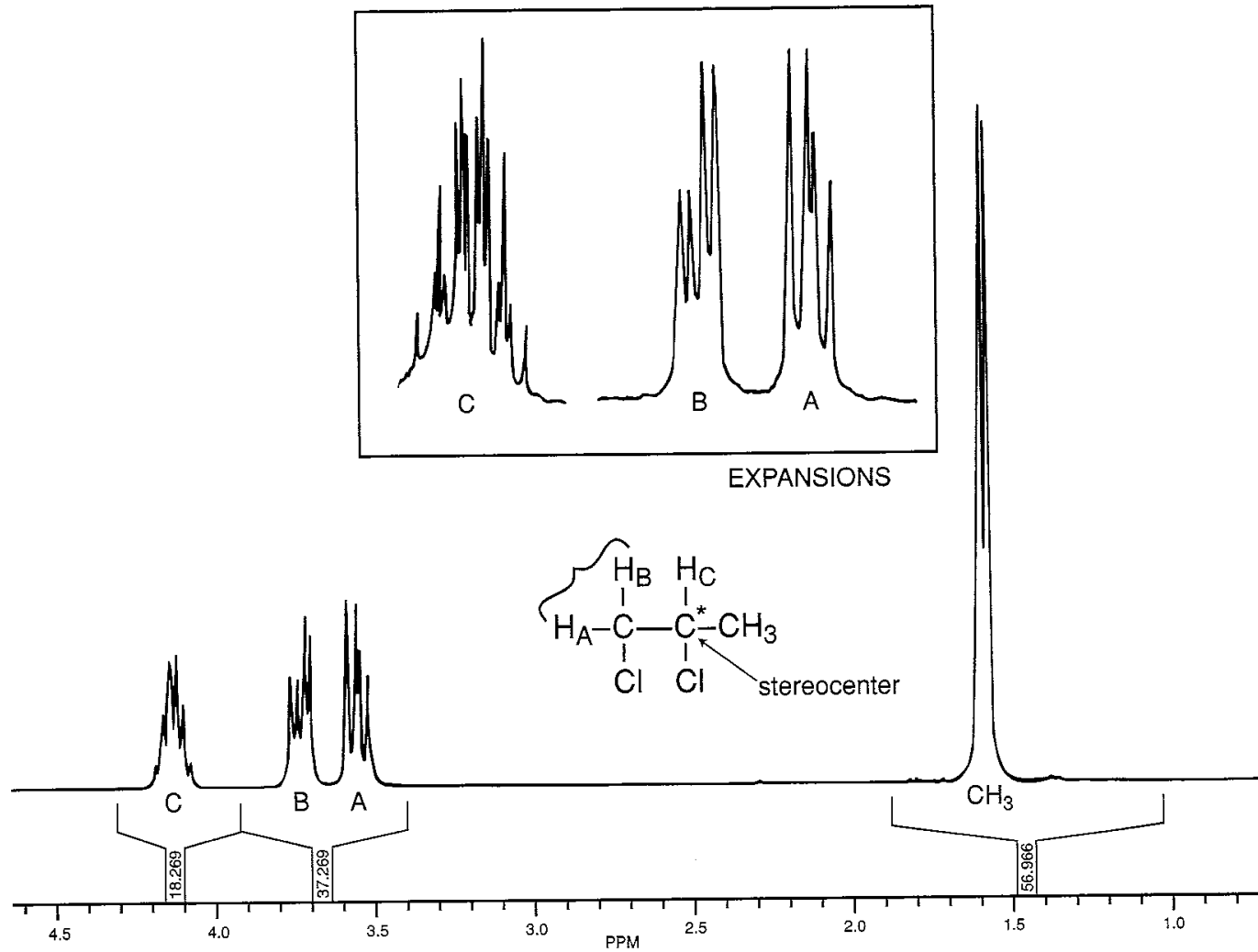


Spin-Spin Coupling: Pascal's Triangle

n	$(n+1)$ multiplicity	relative intensities	multiplet name
0	1	1	singlet
1	2	1:1	doublet
2	3	1:2:1	triplet
3	4	1:3:3:1	quartet
4	5	1:4:6:4:1	quintet
5	6	1:5:10:10:5:1	sextet
6	7	1:6:15:20:15:6:1	septet
7	8	1:7:21:35:35:21:7:1	octet

- This is valid for coupling between spin 1/2 nuclei
- *General rule: $2nI + 1$*

Diastereotopic Protons

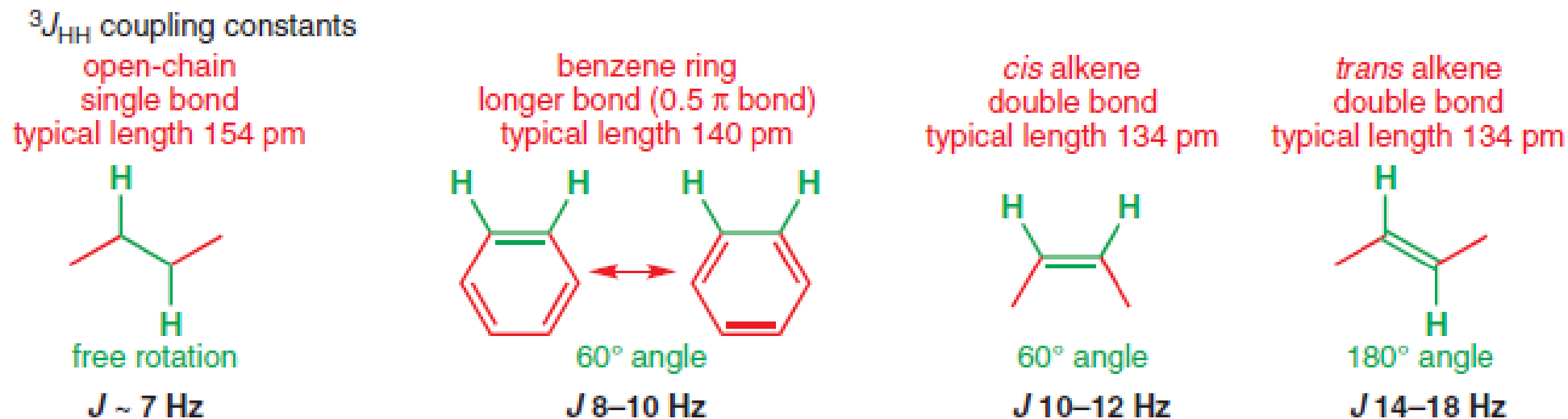


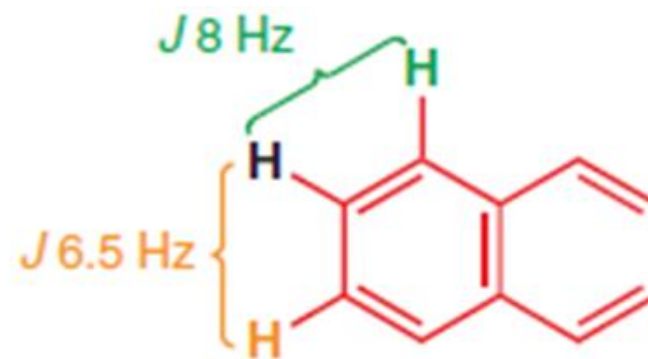
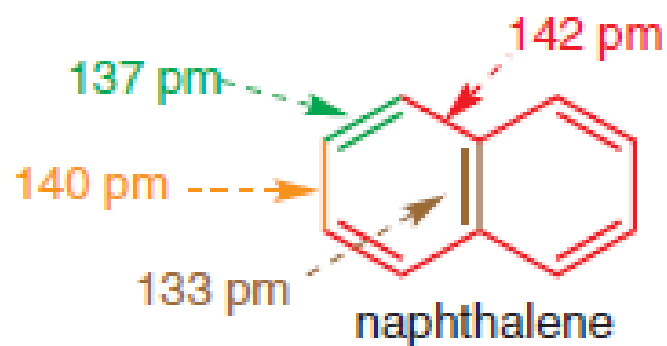
Factors affecting the coupling constant (Vicinal coupling, $^3J_{\text{HH}}$)

- Through-bond distance between the protons.
- Angle between the two C–H bonds.
- Electronegative substituents.

These coupling constants $^3J_{\text{HH}}$ are usually about 7 Hz in an open-chain, freely rotating system. Couplings ($^3J_{\text{HH}}$) across double bonds are usually larger than 7 Hz. $^3J_{\text{HH}}$ couplings are called *vicinal couplings* because the protons concerned are on neighbouring carbon atoms.

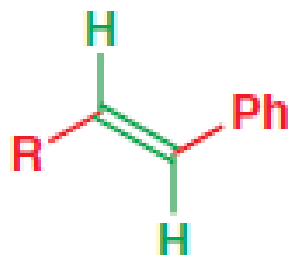
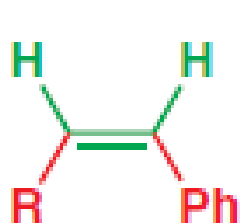
In an open-chain system we have a time average of all rotational conformations. But across a double bond there is no rotation and the angle between the two C–H bonds is fixed: they are always in the same plane. In the plane of the alkene, the C–H bonds are either at 60° (*cis*) or 180° (*trans*) to each other. Coupling constants in benzene rings are slightly less than those across *cis* alkenes because the bond is longer.



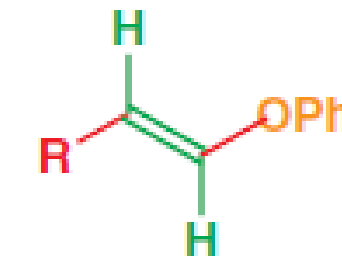
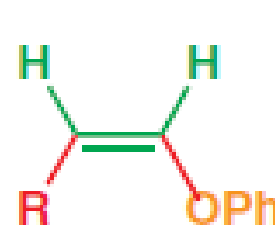


effect of electronegative substituents on $^3J_{HH}$ – alkenes and enol ethers

alkenes



enol ethers



The *trans* coupling for the enol ethers is only just larger than the *cis* coupling for the alkenes.

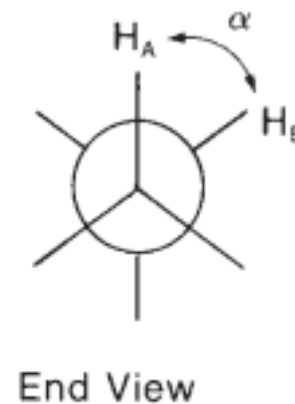
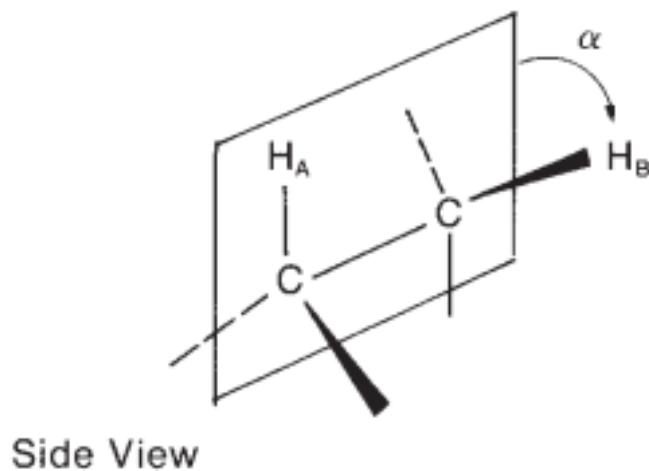
The electronegative oxygen atom is withdrawing electrons from the C–H bond in the enol ethers and weakening communication through the bonds.

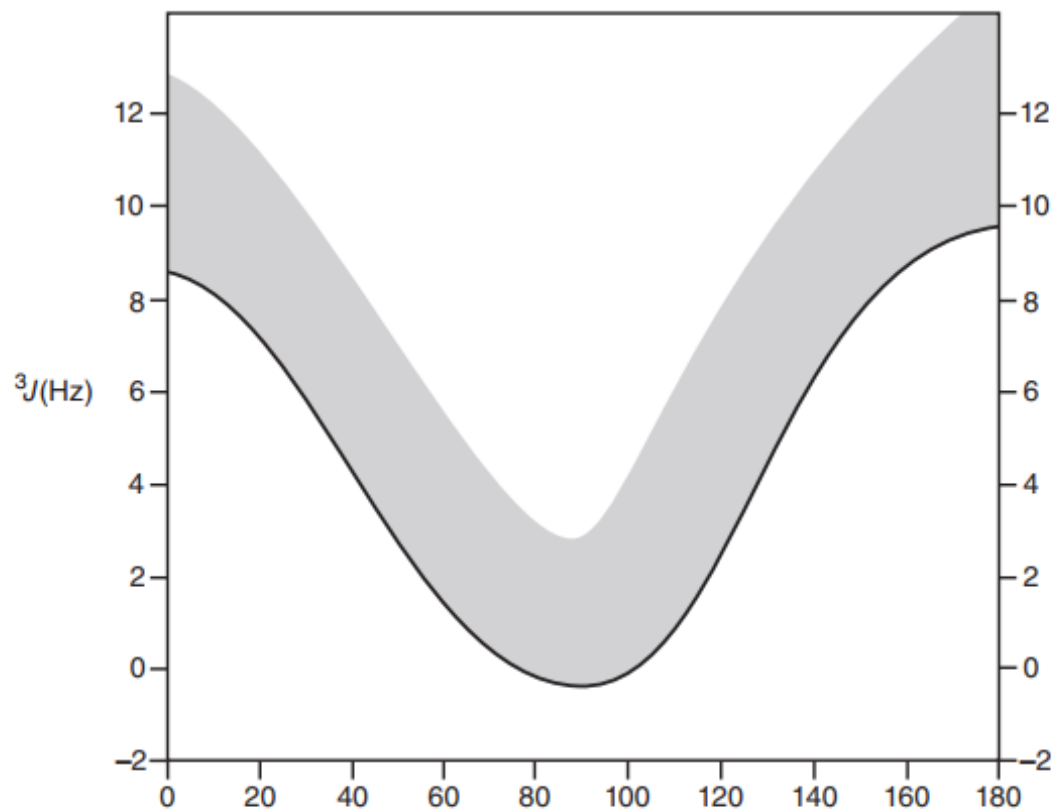
The vicinal coupling constant is the largest when the dihedral angle (angle of rotation, HCCH) is 0° and 180° and the smallest (zero) when the angle is 90° . The side-side overlapping of the two C-H bond orbitals is maximum at 0° , where the C-H orbitals are parallel and is minimum at 90° , where they are perpendicular. At 180° , overlapping with the back lobes of the orbital occurs.

Martin Karplus derived an equation which relates the vicinal coupling constant with the dihedral angle by a mathematical equation as given below.

$$^3J_{\text{HH}} = A + B \cos \alpha + C \cos 2\alpha$$

$$A = 7 \quad B = -1 \quad C = 5$$



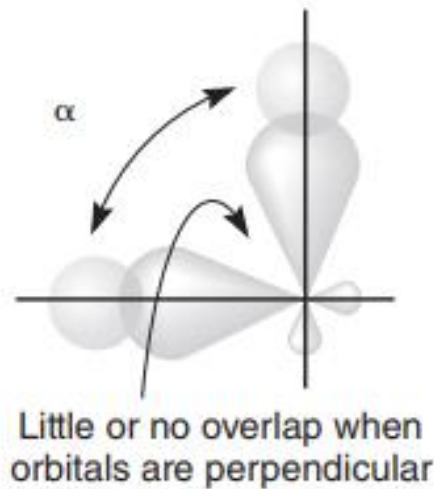


$\alpha = 0^\circ$ (side view)



MAXIMUM OVERLAP

$\alpha = 90^\circ$ (end view)



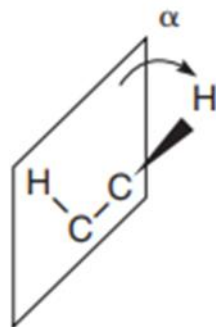
MINIMUM OVERLAP

$\alpha = 180^\circ$ (side view)

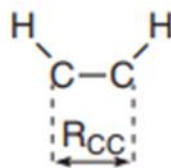


MAXIMUM OVERLAP

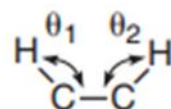
$\alpha \rightarrow$



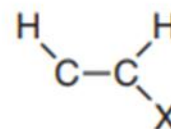
dihedral angle



bond length

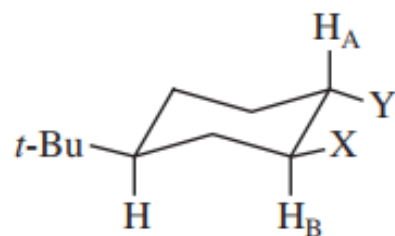


valence angles



electronegative substituents

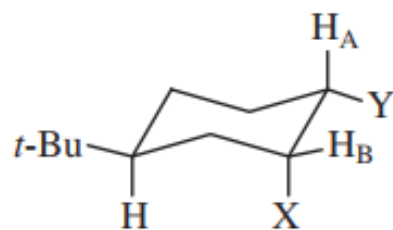
a,a



$$J_{AB} = 10-14 \text{ Hz}$$

$$\alpha = 180^\circ$$

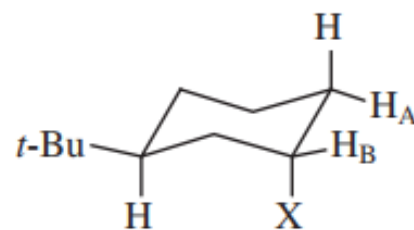
a,e



$$J_{AB} = 2-6 \text{ Hz}$$

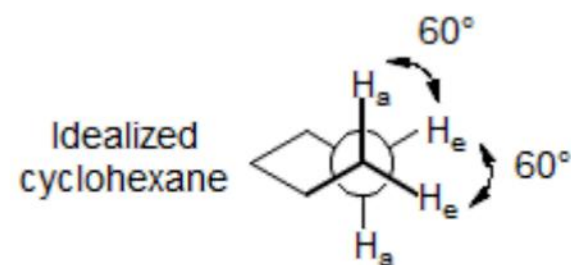
$$\alpha = 60^\circ$$

e,e



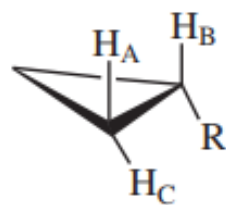
$$J_{AB} = 2-5 \text{ Hz}$$

$$\alpha = 60^\circ$$



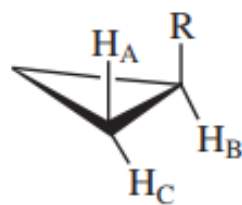
$$^2J_{AC} = 3-9 \text{ Hz}$$

$$\alpha = \sim 115^\circ$$



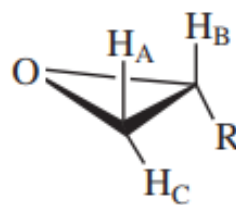
$$^3J_{AB} = 6-12 \text{ Hz}$$

$$\alpha = \sim 0^\circ$$



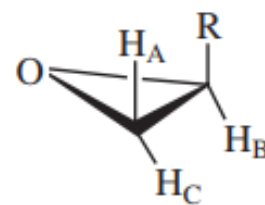
$$^3J_{AB} = 2-9 \text{ Hz}$$

$$\alpha = \sim 120^\circ$$



$$^3J_{AB} = 4-5 \text{ Hz}$$

$$\alpha = \sim 0^\circ$$



$$^3J_{AB} = 2-4 \text{ Hz}$$

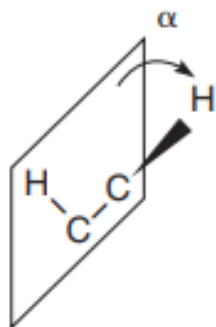
$$\alpha = \sim 120^\circ$$

For three-membered rings, $J_{cis} > J_{trans}$

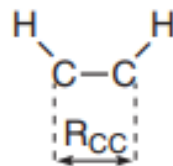
VARIATION OF $^3J_{\text{HH}}$ WITH VALENCE ANGLES IN CYCLIC ALKENES (Hz)				
0-2	2-4	5-7	8-11	6-15

Valence angle (theta)

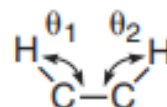
Larger HCH valence angles in the smaller ring sizes result in smaller coupling constants ($^3J_{\text{HH}}$).



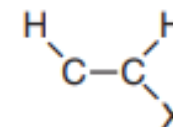
dihedral angle



bond length



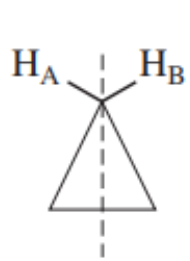
valence angles



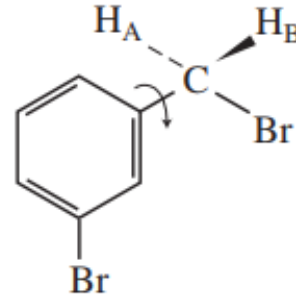
electronegative substituents

Two-Bond Couplings (2J) or geminal couplings

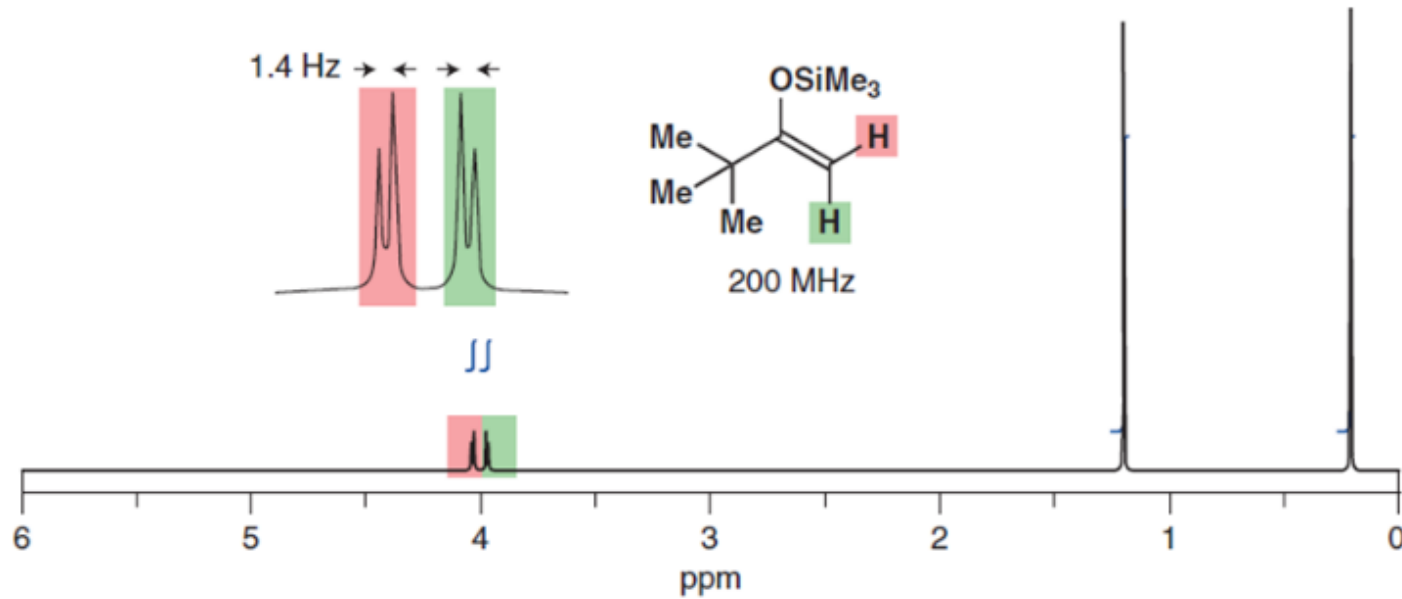
Coupling can occur between protons on the same carbon atom (only if the hydrogens of a CH_2 group are different)

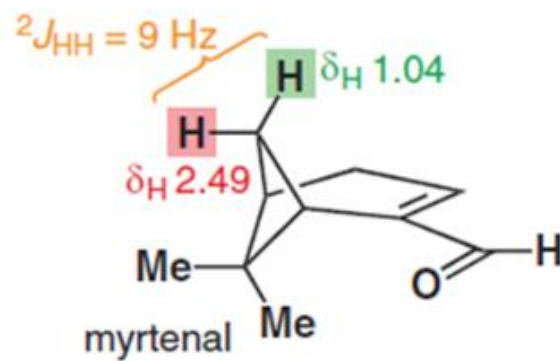
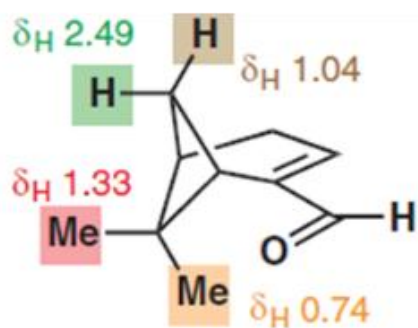
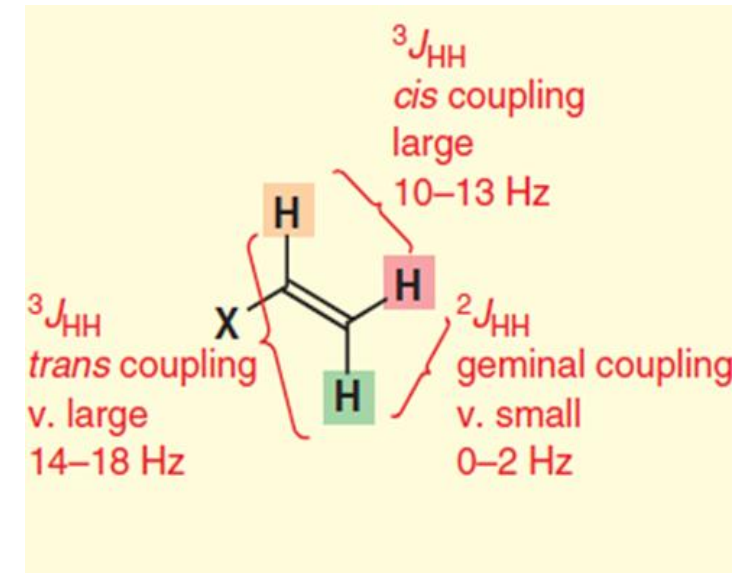
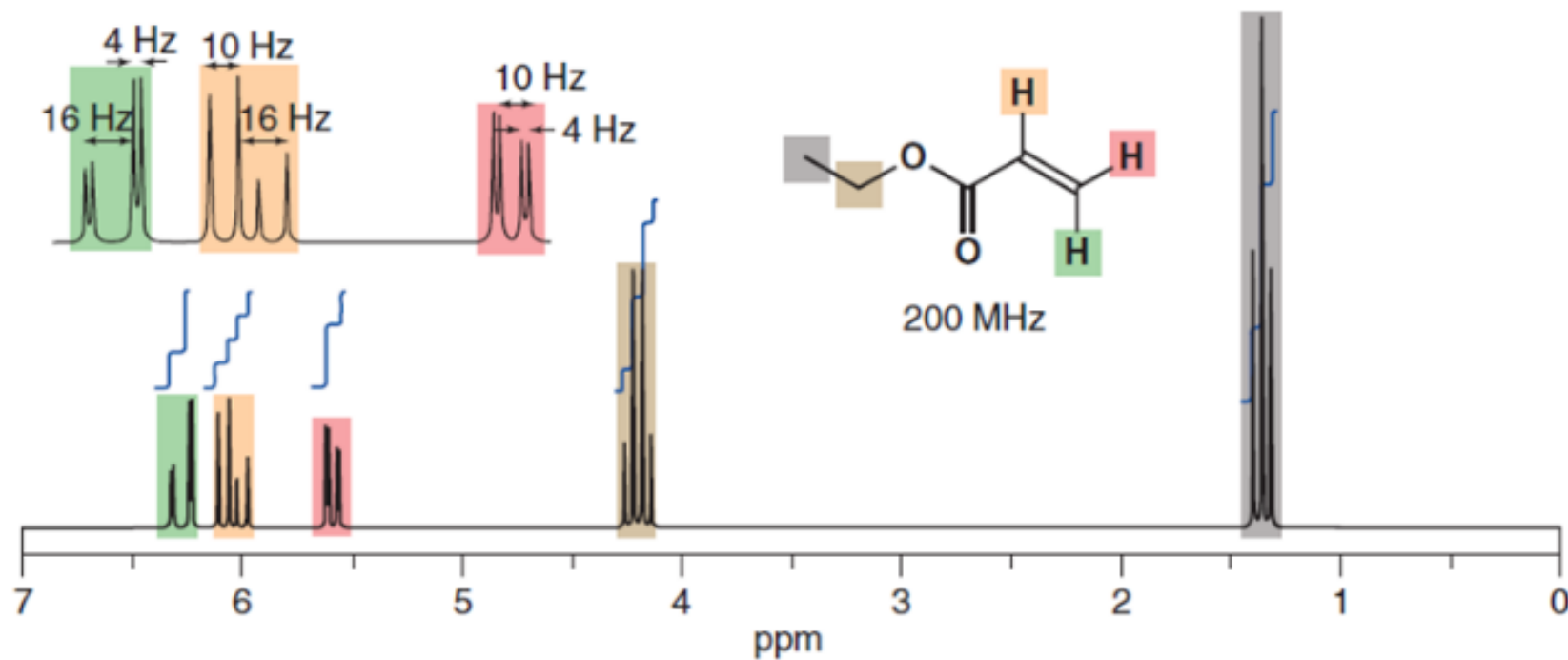


Plane of symmetry—
no splitting



Free rotation—
no splitting

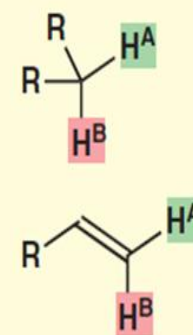




• geminal $^2J_{HH}$

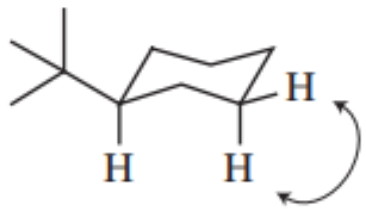
saturated

unsaturated



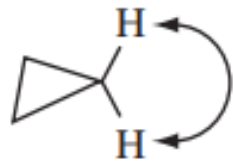
10–16 Hz

0–3 Hz



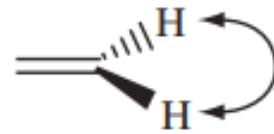
$$\alpha \cong 109^\circ$$

$$^2J_{\text{HH}} \cong 12-18 \text{ Hz}$$



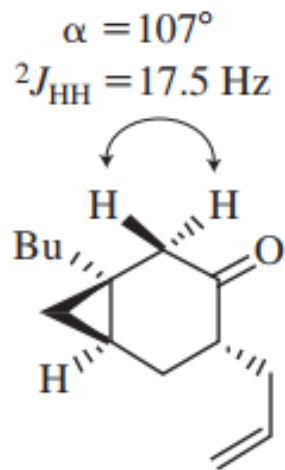
$$\alpha \cong 118^\circ$$

$$^2J_{\text{HH}} \cong 5 \text{ Hz}$$



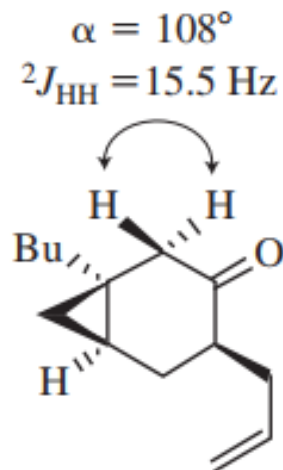
$$\alpha \cong 120^\circ$$

$$^2J_{\text{HH}} \cong 0-3 \text{ Hz}$$



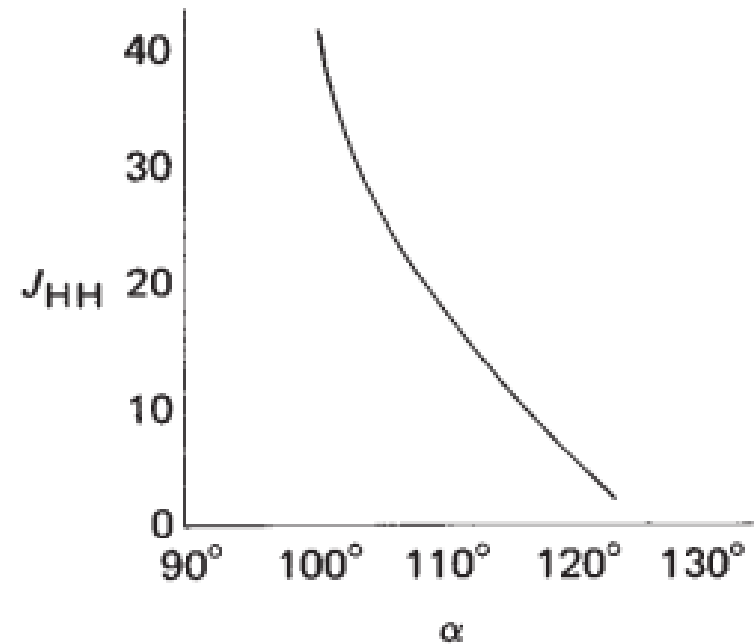
$$\alpha = 107^\circ$$

$$^2J_{\text{HH}} = 17.5 \text{ Hz}$$



$$\alpha = 108^\circ$$

$$^2J_{\text{HH}} = 15.5 \text{ Hz}$$



The amount of geminal coupling depends on the HCH angle α . The graph in RHS Figure shows this dependence, where the amount of *electronic* interaction between the two C-H orbitals determines the magnitude of the coupling constant 2J . In general, 2J geminal coupling constants increase as the angle α decreases. As the angle α decreases, the two orbitals move closer, and the electron spin correlations become greater.

115 deg = 10 Hz

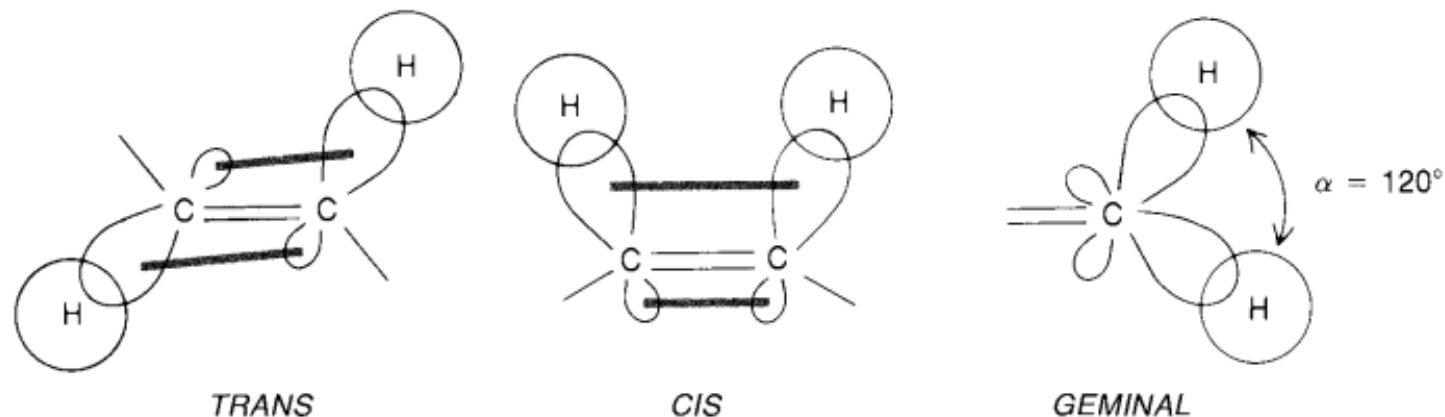
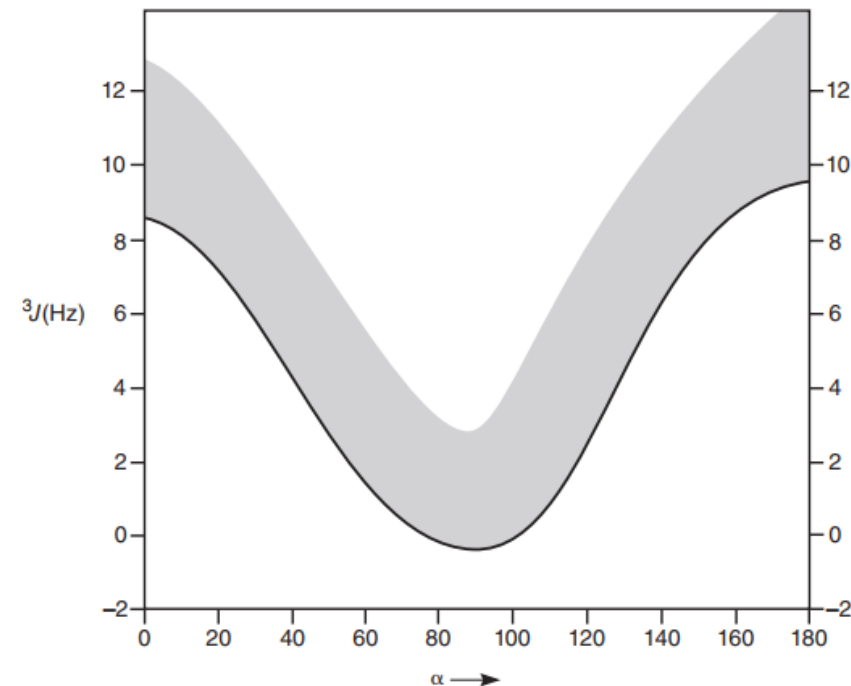
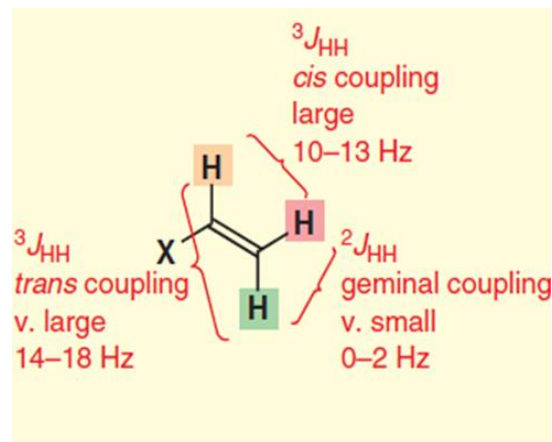
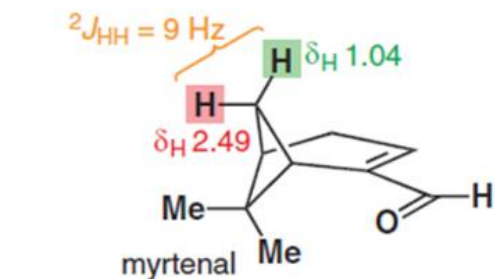
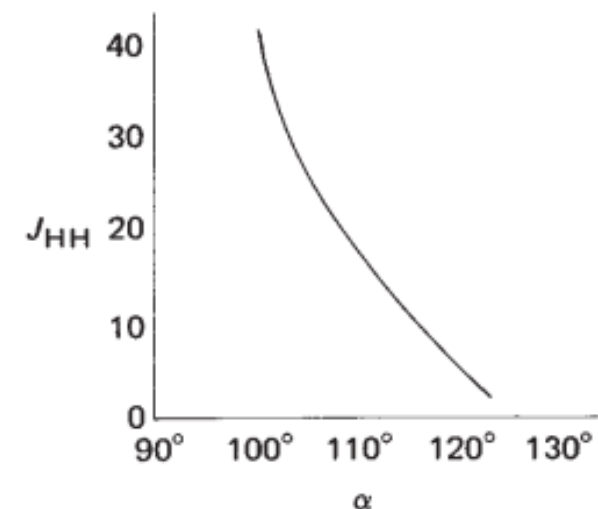


FIGURE 7.52 Coupling mechanisms in alkenes.

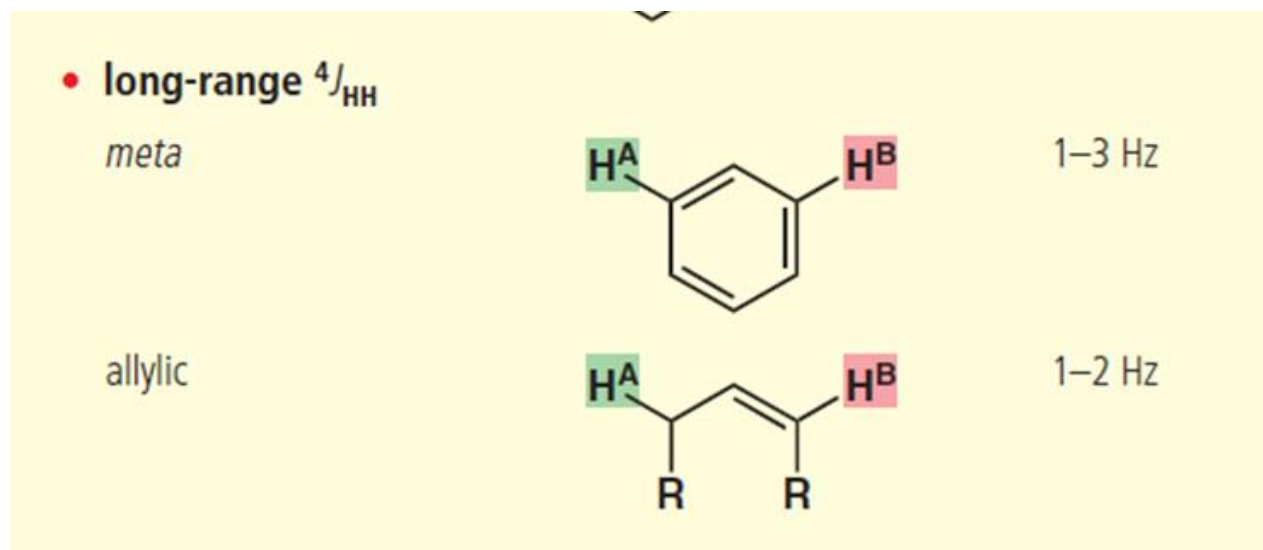
The mechanism of *cis* and *trans* coupling in alkenes is no different from that of any other three-bond vicinal coupling, and that of the terminal methylene protons is just a case of two-bond geminal coupling. All three types have been discussed already and are illustrated in Figure 7.52. To obtain an explanation of the relative magnitudes of the 3J coupling constants, notice that the two C–H bonds are parallel in *trans* coupling, while they are angled away from each other in *cis* coupling. Also note that the H–C–H angle for geminal coupling is nearly 120° , a virtual minimum in the graph of Figure 7.4. In addition to these three types of coupling, alkenes often show small



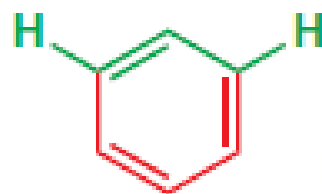
$^2J_{HH}$ vs HCH bond angle (Fig 7.4)

Long-range coupling

When the through-bond distance gets longer than three bonds, coupling is not usually seen. However, it is seen in some special cases, the most important being *meta* coupling in aromatic rings and allylic coupling in alkenes. In both, the orbitals between the two hydrogen atoms can line up in a zig-zag fashion to maximize interaction. This arrangement looks rather like a letter 'W' and this sort of coupling is called W-coupling. Even with this advantage, values of $^4J_{\text{HH}}$ are usually small, about 1–3 Hz.

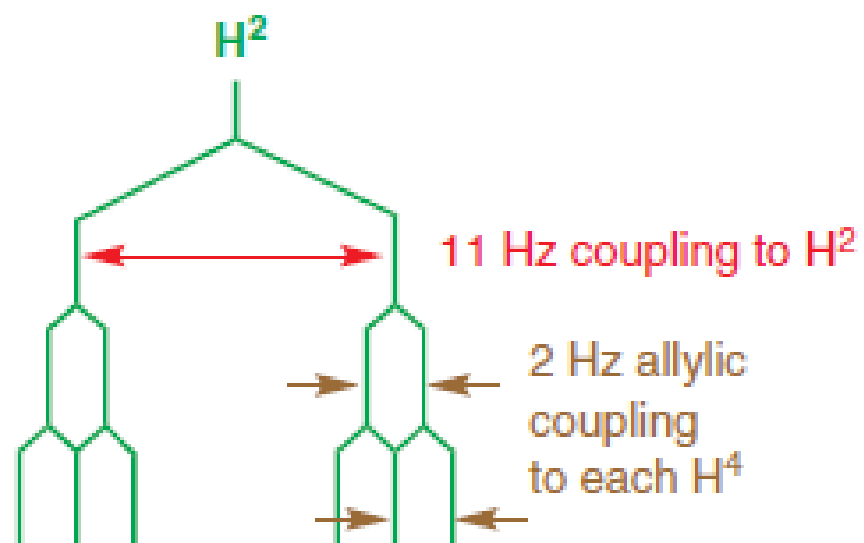
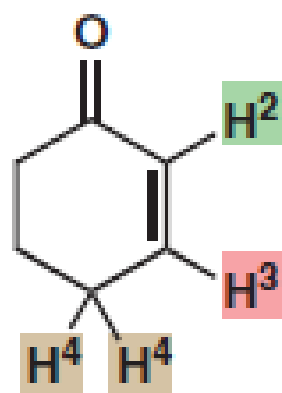
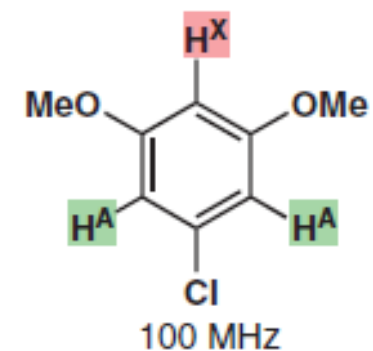
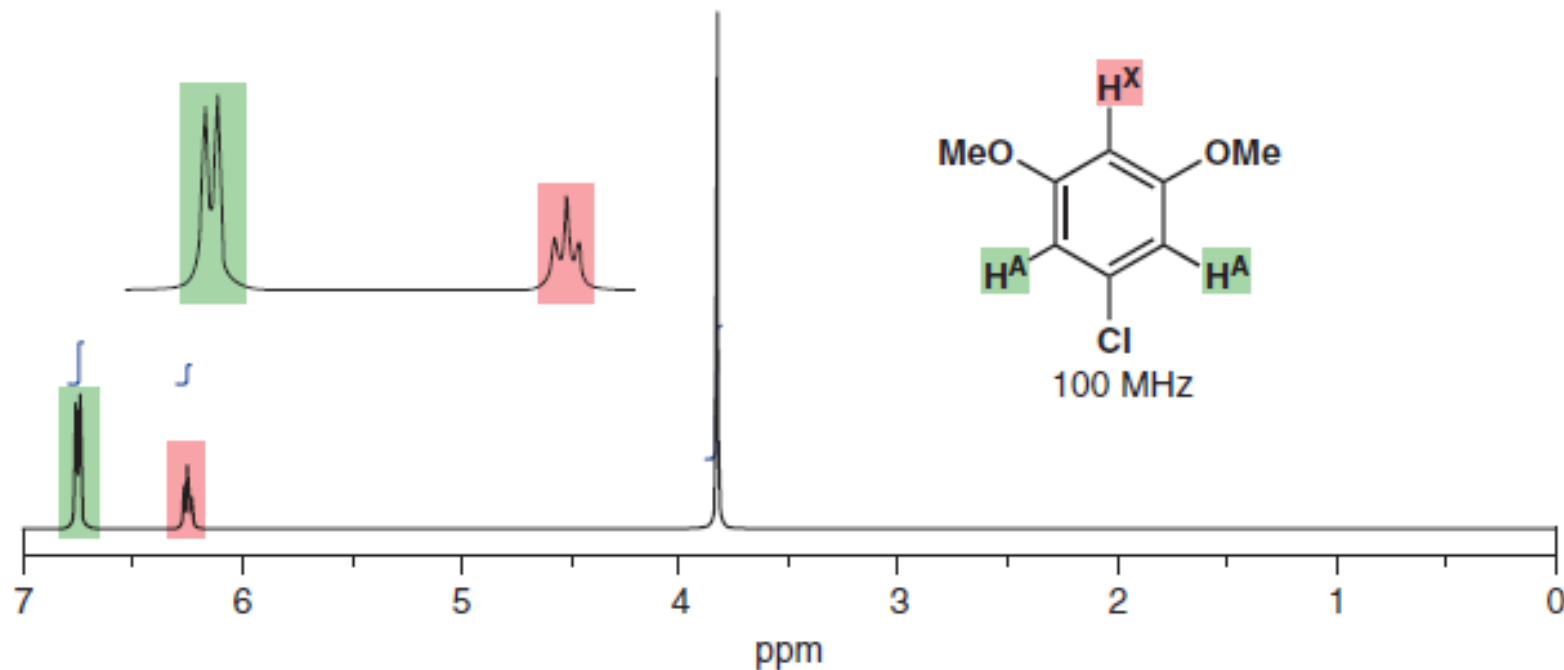
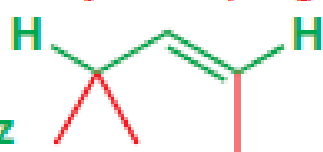


meta coupling



$$0 < {}^4J_{HH} < 3 \text{ Hz}$$

allylic coupling



resultant double triplet for H²:

